Day 2 Activity: Engineering Project Management & Design (Intro Stats)

Name: _____ Date: _____

Scenario: You are a project coordinator for an engineering firm. Your job involves organizing design teams, planning project workflows, and managing technical protocols. Use counting principles to answer each question.

Part 1: Fundamental Counting Principle (FCP)

Show your setup as a product (e.g., choices × choices × choices) and give the final number.

1) Design Team Assignment Assign 1 structural engineer from 4, 1 electrical engineer from 6, and 1 project manager from 3 for a bridge design project. How many different team combinations are possible? Work:

2) Quality Control Pathway A manufactured part must be inspected by 1 of 3 quality inspectors, then tested using 1 of 2 stress test methods, then verified by 1 of 4 certification engineers. How many possible inspection pathways are there? Work:

3) Safety Protocol Sequence An engineer must begin a site inspection with 1 of 2 safety briefings, then use 1 of 3 inspection checklists, and complete 1 of 2 documentation formats. How many protocol sequences exist? Work:

Part 2: Simplifying Factorial Expressions (Numbers Only)

These mirror the calculations inside permutation and combination formulas. Simplify each exactly (cancel before multiplying when possible).

4) $8!/5! =$

5) $10!/(7! \cdot 3!) =$

6) $12!/9! =$

7) $9!/(6! \cdot 3!) =$

8) $15!/(12! \cdot 3!) =$

Part 3: Permutations (Order Matters)

Identify the correct model and compute.

9) Testing Sequence Five prototype designs must be tested in a specific sequence to evaluate performance under different conditions. In how many different orders can the 5 prototypes be tested? Model/Work:

10) Project Leadership Roles From a team of 8 engineers, assign 3 distinct leadership roles: Lead Designer, Quality Assurance Manager, and Safety Officer. In how many ways can these roles be filled? Model/Work:

Part 4: Combinations (Order Does Not Matter)

Identify the correct model and compute.

11) Design Review Committee Select 4 engineers from a group of 10 to serve on the Design Review Committee. How many different committees are possible? Model/Work:

12) Field Site Inspectors Choose 5 engineers from 12 available staff members to conduct field site inspections. How many different groups of 5 can be selected? Model/Work:

Optional Reflection (If time remains) • When should you use FCP instead of permutations or combinations? • How do you recognize when order matters in real engineering contexts (assigned roles vs. work groups)?